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Inventor(s)	Hirukawa; Atsuo et al.

Multilayer coil component

Abstract

A multilayer coil component includes a body including multiple insulating layers stacked in a direction of stacking and having first and second end surfaces opposite each other in a length direction, first and second primary surfaces opposite each other in a height direction, perpendicular to the length direction, and first and second lateral surfaces opposite each other in a width direction, perpendicular to the length direction and to the height direction; a coil inside the body and including multiple coil conductors electrically connected together; and a first outer electrode extending from at least part of the first end surface of the body to part of the first primary surface and electrically coupled to the coil. The direction of stacking of the insulating layers and the direction of the coil axis of the coil are parallel with the first primary surface, which is the mounting surface, of the body.

Inventors: Hirukawa; Atsuo (Nagaokakyo, JP), Nohchai; Suradech (Nagaokakyo, JP), Takai; Shun (Nagaokakyo, JP), Koshiji; Kenjirou (Nagaokakyo, JP)

Applicant: Murata Manufacturing Co., Ltd. (Kyoto-fu, JP)

Family ID: 82740193

Assignee: Murata Manufacturing Co., Ltd. (Kyoto-fu, JP)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2012/0188682	12/2011	Sato	336/200	H01G 4/232
2017/0345552	12/2016	Nakano	N/A	H01F 41/043
2019/0148060	12/2018	Arai	336/192	H01F 27/2804
2019/0198232	12/2018	Sato	N/A	H01F 27/2804

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102610388	12/2011	CN	N/A
111653412	12/2019	CN	N/A
2014-212298	12/2013	JP	N/A
2015-109410	12/2014	JP	N/A
2017-212372	12/2016	JP	N/A
2018-170429	12/2017	JP	N/A
2019-186255	12/2018	JP	N/A
2020-145223	12/2019	JP	N/A

Primary Examiner: Hinson; Ronald

Attorney, Agent or Firm: Studebaker Brackett PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims benefit of priority to Japanese Patent Application No. 2021-023406, filed Feb. 17, 2021, the entire content of which is incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to a multilayer coil component.

Background Art

(3) As a multilayer coil component superior in radiofrequency characteristics, Japanese Unexamined Patent Application Publication No. 2019-186255 discloses a multilayer coil component having a transmission coefficient S_{21} at 40 GHz of -1.0 dB or more and 0 dB or less (i.e., from -1.0 dB to 0 dB).

SUMMARY

(4) As high-speed and large-capacity communication has advanced, however, multilayer coil components are now expected to have a high transmission coefficient S_{21} over higher-frequency bands and be consistent in the transmission coefficient S_{21} in the radiofrequency and lower-frequency bands at the same time.

(5) Accordingly, the present disclosure provides a multilayer coil component that has a high and consistent transmission coefficient S_{21} in the radiofrequency band.

(6) The multilayer coil component according to the present disclosure includes a body formed by a plurality of insulating layers stacked in a direction of stacking and having first and second end surfaces opposite each other in a length direction, first and second primary surfaces opposite each other in a height direction, perpendicular to the length direction, and first and second lateral surfaces opposite each other in a width direction, perpendicular to the length direction and to the height direction. The multilayer coil component further includes a coil disposed inside the body and formed by a plurality of coil conductors electrically connected together; and a first outer electrode extending from at least part of the first end surface of the body to part of the first primary surface and electrically coupled to the coil. The direction of stacking of the insulating layers and a direction of a coil axis of the coil are parallel with the first primary surface, which is a mounting surface, of the body. At least part of the body has a magnetic phase containing Fe, Ni, Zn, and Cu and a nonmagnetic phase containing Si. The first outer electrode has, in order from a body side, an underlying electrode and a plating electrode on the underlying electrode; an end in the length direction of a portion of the plating electrode lying on the first primary surface of the body is closer to the second end surface of the body than an end in the length direction of a portion of the underlying electrode lying on the first primary surface of the body; and a distance in the length direction between the end of the plating electrode and the end of the underlying electrode is $30\text{ }\mu\text{m}$ or less.

(7) According to the present disclosure, there can be provided a multilayer coil component that has a high and consistent transmission coefficient S_{21} in the radiofrequency band.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective schematic diagram illustrating an example of a multilayer coil component according to the present disclosure;

(2) FIG. 2 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the first end surface side of the body;

(3) FIG. 3 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the first primary surface side of the body;

(4) FIG. 4 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the first lateral surface side of the body;

(5) FIG. 5 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the second lateral surface side of the body;

(6) FIG. 6 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the second end surface side of the body;

(7) FIG. 7 is a cross-sectional schematic diagram illustrating an example of a portion corresponding

to line segment A1-A2 in FIG. 1;

(8) FIG. 8 is a perspective schematic diagram illustrating an example of a disassembled form of the body and coil in FIG. 7;

(9) FIG. 9 is a plan schematic diagram illustrating an example of a disassembled form of the body and coil in FIG. 7;

(10) FIG. 10 is a side schematic view of the multilayer coil component in FIG. 1;

(11) FIG. 11 is a schematic diagram illustrating part of a cross-section of the first body section and first outer electrode in FIG. 10 parallel with the length and height directions;

(12) FIG. 12 is a schematic diagram illustrating part of a cross-section of the second body section and second outer electrode in FIG. 10 parallel with the length and height directions; and

(13) FIG. 13 is a graphical representation of measured transmission coefficients S21 of multilayer coil component samples 1, 3, and 4.

DETAILED DESCRIPTION

(14) The following describes the multilayer coil component according to the present disclosure. It should be noted that the present disclosure is not limited to the following configurations and may optionally be modified within the scope of the present disclosure. Combinations of multiple ones of the specific preferred configurations set forth below are also the present disclosure.

(15) The multilayer coil component according to the present disclosure includes a body formed by multiple insulating layers stacked in a direction of stacking and having first and second end surfaces opposite each other in a length direction, first and second primary surfaces opposite each other in a height direction, perpendicular to the length direction, and first and second lateral surfaces opposite each other in a width direction, perpendicular to the length direction and to the height direction. The multilayer coil component further includes a coil disposed inside the body and formed by multiple coil conductors electrically connected together; and a first outer electrode extending from at least part of the first end surface of the body to part of the first primary surface and electrically coupled to the coil.

(16) FIG. 1 is a perspective schematic diagram illustrating an example of a multilayer coil component according to the present disclosure.

(17) As illustrated in FIG. 1, the multilayer coil component 1 has a body 10, a first outer electrode 21, and a second outer electrode 22. Although not illustrated in FIG. 1, the multilayer coil component 1 also has, as described later herein, a coil disposed inside the body 10.

(18) As illustrated, for example in FIG. 1, length, height, and width directions are herein defined as the directions denoted by L, T, and W, respectively. The length direction L, the height direction T, and the width direction W are perpendicular to one another.

(19) The body 10 has a first end surface 11a and a second end surface 11b opposite each other in the length direction L, a first primary surface 12a and a second primary surface 12b opposite each other in the height direction T, and a first lateral surface 13a and a second lateral surface 13b opposite each other in the width direction W and is, for example, cuboid or substantially cuboid in shape.

(20) The first and second end surfaces 11a and 11b of the body 10 do not need to be strictly perpendicular to the length direction L. The first and second primary surfaces 12a and 12b of the body 10 do not need to be strictly perpendicular to the height direction T either. The first and second lateral surfaces 13a and 13b of the body 10, furthermore, do not need to be strictly perpendicular to the width direction W.

(21) When the multilayer coil component 1 is mounted onto a substrate, the first primary surface 12a of the body 10 is the mounting surface.

(22) Preferably, the body 10 has rounded corners and edges. A corner of the body 10 is a point of the body 10 at which three of its surfaces meet. An edge of the body 10 is a portion of the body 10 at which two of its surfaces meet.

(23) FIG. 2 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when

viewed from the first end surface side of the body. FIG. 3 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the first primary surface side of the body. FIG. 4 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the first lateral surface side of the body. FIG. 5 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the second lateral surface side of the body. FIG. 6 is a plan schematic diagram illustrating the multilayer coil component in FIG. 1 when viewed from the second end surface side of the body.

(24) As illustrated in FIGS. 1, 2, and 3, the first outer electrode **21** extends from at least part of the first end surface **11a** of the body **10**, from part of the first end surface **11a** of the body **10** here, to part of the first primary surface **12a**. The presence of the first outer electrode **21** on the first primary surface **12a**, which is the mounting surface, of the body **10**, improves the ease of mounting of the multilayer coil component **1**.

(25) As illustrated in FIG. 2, the first outer electrode **21** covers a region of the first end surface **11a** of the body **10** including the edge at which it meets the first primary surface **12a** and does not cover a region including the edge at which it meets the second primary surface **12b**. This makes the first end surface **11a** of the body **10** exposed in its region including the edge at which it meets the second primary surface **12b**.

(26) When the first outer electrode **21** is viewed in the length direction **L**, its dimension **E.sub.1** in the height direction **T** does not need to be constant in the width direction **W**, although it is constant in FIG. 2. For example, when viewed in the length direction **L**, the first outer electrode **21** may have an arched shape, in which the dimension **E.sub.1** in the height direction **T** increases from the ends toward the middle in the width direction **W**.

(27) The first outer electrode **21** may be on part of the first end surface **11a** of the body **10** as illustrated in FIGS. 1 and 2 or may be over the entire first end surface **11a** of the body **10**.

(28) As illustrated in FIG. 3, the first outer electrode **21** covers a region of the first primary surface **12a** of the body **10** including the edge at which it meets the first end surface **11a** and does not cover a region including the edge at which it meets the second end surface **11b**.

(29) When the first outer electrode **21** is viewed in the height direction **T**, its dimension **E.sub.2** in the length direction **L** does not need to be constant in the width direction **W**, although it is constant in FIG. 3. For example, when viewed in the height direction **T**, the first outer electrode **21** may have an arched shape, in which the dimension **E.sub.2** in the length direction **L** increases from the ends toward the middle in the width direction **W**.

(30) As illustrated in FIGS. 1, 4, and 5, the first outer electrode **21** may extend from part of the first end surface **11a** of the body **10** to part of the first primary surface **12a** and also to part of the first lateral surface **13a** and part of the second lateral surface **13b**. More specifically, the first outer electrode **21** may cover a region of the first lateral surface **13a** of the body **10** including the vertex at which it meets the first end surface **11a** and the first primary surface **12a** without covering a region including the vertex at which it meets the first end surface **11a** and the second primary surface **12b**. The first outer electrode **21**, furthermore, may cover a region of the second lateral surface **13b** of the body **10** including the vertex at which it meets the first end surface **11a** and the first primary surface **12a** without covering a region including the vertex at which it meets the first end surface **11a** and the second primary surface **12b**.

(31) Preferably, as illustrated in FIG. 4, the contours of the portion of the first outer electrode **21** covering the first lateral surface **13a** of the body **10** include a first contour **50a** facing the edge at which the first lateral surface **13a** and the first end surface **11a** meet, a second contour **50b** facing the edge at which the first lateral surface **13a** and the first primary surface **12a** meet, and a contour diagonal to the first and second contours **50a** and **50b** in addition to these.

(32) Preferably, as illustrated in FIG. 5, the contours of the portion of the first outer electrode **21** covering the second lateral surface **13b** of the body **10** include a third contour **50c** facing the edge at which the second lateral surface **13b** and the first end surface **11a** meet, a fourth contour **50d**

facing the edge at which the second lateral surface **13b** and the first primary surface **12a** meet, and a contour diagonal to the third and fourth contours **50c** and **50d** in addition to these.

(33) The first outer electrode **21** may be off the first lateral surface **13a** of the body **10**. The first outer electrode **21**, furthermore, may be off the second lateral surface **13b** of the body **10**.

(34) The first outer electrode **21** may extend from the first end surface **11a** of the body **10** to part of each of the first primary surface **12a**, second primary surface **12b**, first lateral surface **13a**, and second lateral surface **13b**.

(35) As illustrated in FIGS. **1**, **3**, and **6**, the second outer electrode **22** extends from at least part of the second end surface **11b** of the body **10**, from part of the second end surface **11b** of the body **10** here, to part of the first primary surface **12a**. The presence of the second outer electrode **22** on the first primary surface **12a**, which is the mounting surface, of the body **10**, improves the ease of mounting of the multilayer coil component **1**.

(36) As illustrated in FIG. **6**, the second outer electrode **22** covers a region of the second end surface **11b** of the body **10** including the edge at which it meets the first primary surface **12a** and does not cover a region including the edge at which it meets the second primary surface **12b**. This makes the second end surface **11b** of the body **10** exposed in its region including the edge at which it meets the second primary surface **12b**.

(37) When the second outer electrode **22** is viewed in the length direction **L**, its dimension **E.sub.3** in the height direction **T** does not need to be constant in the width direction **W**, although it is constant in FIG. **6**. For example, when viewed in the length direction **L**, the second outer electrode **22** may have an arched shape, in which the dimension **E.sub.3** in the height direction **T** increases from the ends toward the middle in the width direction **W**.

(38) The second outer electrode **22** may be on part of the second end surface **11b** of the body **10** as illustrated in FIGS. **1** and **6** or may be over the entire second end surface **11b** of the body **10**.

(39) As illustrated in FIG. **3**, the second outer electrode **22** covers a region of the first primary surface **12a** of the body **10** including the edge at which it meets the second end surface **11b** and does not cover a region including the edge at which it meets the first end surface **11a**.

(40) When the second outer electrode **22** is viewed in the height direction **T**, its dimension **E.sub.4** in the length direction **L** does not need to be constant in the width direction **W**, although it is constant in FIG. **3**. For example, when viewed in the height direction **T**, the second outer electrode **22** may have an arched shape, in which the dimension **E.sub.4** in the length direction **L** increases from the ends toward the middle in the width direction **W**.

(41) As illustrated in FIGS. **1**, **4**, and **5**, the second outer electrode **22** may extend from part of the second end surface **11b** of the body **10** to part of the first primary surface **12a** and also to part of the first lateral surface **13a** and part of the second lateral surface **13b**. More specifically, the second outer electrode **22** may cover a region of the first lateral surface **13a** of the body **10** including the vertex at which it meets the second end surface **11b** and the first primary surface **12a** without covering a region including the vertex at which it meets the second end surface **11b** and the second primary surface **12b**. The second outer electrode **22**, furthermore, may cover a region of the second lateral surface **13b** of the body **10** including the vertex at which it meets the second end surface **11b** and the first primary surface **12a** without covering a region including the vertex at which it meets the second end surface **11b** and the second primary surface **12b**.

(42) Preferably, as illustrated in FIG. **4**, the contours of the portion of the second outer electrode **22** covering the first lateral surface **13a** of the body **10** include a fifth contour **50e** facing the edge at which the first lateral surface **13a** and the second end surface **11b** meet, a sixth contour **50f** facing the edge at which the first lateral surface **13a** and the first primary surface **12a** meet, and a contour diagonal to the fifth and sixth contours **50e** and **50f** in addition to these.

(43) Preferably, as illustrated in FIG. **5**, the contours of the portion of the second outer electrode **22** covering the second lateral surface **13b** of the body **10** include a seventh contour **50g** facing the edge at which the second lateral surface **13b** and the second end surface **11b** meet, an eighth

contour **50h** facing the edge at which the second lateral surface **13b** and the first primary surface **12a** meet, and a contour diagonal to the seventh and eighth contours **50g** and **50h** in addition to these.

(44) The second outer electrode **22** may be off the first lateral surface **13a** of the body **10**. The second outer electrode **22**, furthermore, may be off the second lateral surface **13b** of the body **10**.

(45) The second outer electrode **22** may extend from the second end surface **11b** of the body **10** to part of each of the first primary surface **12a**, second primary surface **12b**, first lateral surface **13a**, and second lateral surface **13b**.

(46) The size of the multilayer coil component **1** is not critical, but preferably is the 0603, 0402, or 1005 size.

(47) For when the multilayer coil component **1** is the 0603, 0402, or 1005 size, the following presents specific examples of preferred dimensions of the multilayer coil component **1**, body **10**, first outer electrode **21**, and second outer electrode **22**.

(48) (1) When the multilayer coil component **1** is the 0603 size Preferably, the dimension L.sub.1 of the multilayer coil component **1** in the length direction L is 0.57 mm or more. Preferably, furthermore, the dimension L.sub.1 of the multilayer coil component **1** in the length direction L is 0.63 mm or less. Preferably, the dimension T.sub.1 of the multilayer coil component **1** in the height direction T is 0.27 mm or more. Preferably, furthermore, the dimension T.sub.1 of the multilayer coil component **1** in the height direction T is 0.33 mm or less. Preferably, the dimension W.sub.1 of the multilayer coil component **1** in the width direction W is 0.27 mm or more. Preferably, furthermore, the dimension W.sub.1 of the multilayer coil component **1** in the width direction W is 0.33 mm or less. Preferably, the dimension L.sub.2 of the body **10** in the length direction L is 0.57 mm or more. Preferably, furthermore, the dimension L.sub.2 of the body **10** in the length direction L is 0.63 mm or less. Preferably, the dimension T.sub.2 of the body **10** in the height direction T is 0.27 mm or more. Preferably, furthermore, the dimension T.sub.2 of the body **10** in the height direction T is 0.33 mm or less. Preferably, the dimension W.sub.2 of the body **10** in the width direction W is 0.27 mm or more. Preferably, furthermore, the dimension W.sub.2 of the body **10** in the width direction W is 0.33 mm or less. Preferably, the dimension E.sub.1 of the first outer electrode **21** in the height direction T is 0.10 mm or more and 0.20 mm or less (i.e., from 0.10 mm to 0.20 mm). If the dimension E.sub.1 of the first outer electrode **21** in the height direction T is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.2 of the first outer electrode **21** in the length direction L is 0.12 mm or more and 0.22 mm or less (i.e., from 0.12 mm to 0.22 mm). If the dimension E.sub.2 of the first outer electrode **21** in the length direction L is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.3 of the second outer electrode **22** in the height direction T is 0.10 mm or more and 0.20 mm or less (i.e., from 0.10 mm to 0.20 mm). If the dimension E.sub.3 of the second outer electrode **22** in the height direction T is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.4 of the second outer electrode **22** in the length direction L is 0.12 mm or more and 0.22 mm or less (i.e., from 0.12 mm to 0.22 mm). If the dimension E.sub.4 of the second outer electrode **22** in the length direction L is not constant in the width direction W, it is preferred that the maximum be in this range.

(49) (2) When the multilayer coil component **1** is the 0402 size Preferably, the dimension L.sub.1 of the multilayer coil component **1** in the length direction L is 0.38 mm or more. Preferably, furthermore, the dimension L.sub.1 of the multilayer coil component **1** in the length direction L is 0.42 mm or less. Preferably, the dimension T.sub.1 of the multilayer coil component **1** in the height direction T is 0.18 mm or more. Preferably, furthermore, the dimension T.sub.1 of the multilayer coil component **1** in the height direction T is 0.22 mm or less. Preferably, the dimension W.sub.1 of the multilayer coil component **1** in the width direction W is 0.18 mm or more. Preferably, furthermore, the dimension W.sub.1 of the multilayer coil component **1** in the width direction W is

0.22 mm or less. Preferably, the dimension L.sub.2 of the body **10** in the length direction L is 0.38 mm or more. Preferably, furthermore, the dimension L.sub.2 of the body **10** in the length direction L is 0.42 mm or less. Preferably, the dimension T.sub.2 of the body **10** in the height direction T is 0.18 mm or more. Preferably, furthermore, the dimension T.sub.2 of the body **10** in the height direction T is 0.22 mm or less. Preferably, the dimension W.sub.2 of the body **10** in the width direction W is 0.18 mm or more. Preferably, furthermore, the dimension W.sub.2 of the body **10** in the width direction W is 0.22 mm or less. Preferably, the dimension E.sub.1 of the first outer electrode **21** in the height direction T is 0.06 mm or more and 0.13 mm or less (i.e., from 0.06 mm to 0.13 mm). If the dimension E.sub.1 of the first outer electrode **21** in the height direction T is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.2 of the first outer electrode **21** in the length direction L is 0.08 mm or more and 0.15 mm or less (i.e., from 0.08 mm to 0.15 mm). If the dimension E.sub.2 of the first outer electrode **21** in the length direction L is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.3 of the second outer electrode **22** in the height direction T is 0.06 mm or more and 0.13 mm or less (i.e., from 0.06 mm to 0.13 mm). If the dimension E.sub.3 of the second outer electrode **22** in the height direction T is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.4 of the second outer electrode **22** in the length direction L is 0.08 mm or more and 0.15 mm or less (i.e., from 0.08 mm to 0.15 mm). If the dimension E.sub.4 of the second outer electrode **22** in the length direction L is not constant in the width direction W, it is preferred that the maximum be in this range.

(50) (3) When the multilayer coil component **1** is the 1005 size Preferably, the dimension L.sub.1 of the multilayer coil component **1** in the length direction L is 0.95 mm or more. Preferably, furthermore, the dimension L.sub.1 of the multilayer coil component **1** in the length direction L is 1.05 mm or less. Preferably, the dimension T.sub.1 of the multilayer coil component **1** in the height direction T is 0.45 mm or more. Preferably, furthermore, the dimension T.sub.1 of the multilayer coil component **1** in the height direction T is 0.55 mm or less. Preferably, the dimension W.sub.1 of the multilayer coil component **1** in the width direction W is 0.45 mm or more. Preferably, furthermore, the dimension W.sub.1 of the multilayer coil component **1** in the width direction W is 0.55 mm or less. Preferably, the dimension L.sub.2 of the body **10** in the length direction L is 0.95 mm or more. Preferably, furthermore, the dimension L.sub.2 of the body **10** in the length direction L is 1.05 mm or less. Preferably, the dimension T.sub.2 of the body **10** in the height direction T is 0.45 mm or more. Preferably, furthermore, the dimension T.sub.2 of the body **10** in the height direction T is 0.55 mm or less. Preferably, the dimension W.sub.2 of the body **10** in the width direction W is 0.45 mm or more. Preferably, furthermore, the dimension W.sub.2 of the body **10** in the width direction W is 0.55 mm or less. Preferably, the dimension E.sub.1 of the first outer electrode **21** in the height direction T is 0.15 mm or more and 0.33 mm or less (i.e., from 0.15 mm to 0.33 mm). If the dimension E.sub.1 of the first outer electrode **21** in the height direction T is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.2 of the first outer electrode **21** in the length direction L is 0.20 mm or more and 0.38 mm or less (i.e., from 0.20 mm to 0.38 mm). If the dimension E.sub.2 of the first outer electrode **21** in the length direction L is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.3 of the second outer electrode **22** in the height direction T is 0.15 mm or more and 0.33 mm or less (i.e., from 0.15 mm to 0.33 mm). If the dimension E.sub.3 of the second outer electrode **22** in the height direction T is not constant in the width direction W, it is preferred that the maximum be in this range. Preferably, the dimension E.sub.4 of the second outer electrode **22** in the length direction L is 0.20 mm or more and 0.38 mm or less (i.e., from 0.20 mm to 0.38 mm). If the dimension E.sub.4 of the second outer electrode **22** in the length direction L is not constant in the width direction W, it is preferred that the maximum be in this range.

(51) For the multilayer coil component according to the present disclosure, the direction of stacking of the insulating layers and the direction of the coil axis of the coil are parallel with the first primary surface, which is the mounting surface, of the body.

(52) FIG. 7 is a cross-sectional schematic diagram illustrating an example of a portion corresponding to line segment A1-A2 in FIG. 1.

(53) As illustrated in FIG. 7, the body **10** is formed by multiple insulating layers **15** stacked in a direction of stacking, in the length direction L here. That is, the direction of stacking of the insulating layers **15** is parallel with the length direction L and parallel with the first primary surface **12a**, which is the mounting surface, of the body **10**. It should be noted that in FIG. 7, the insulating layers **15** are illustrated with lines therebetween for the convenience of description, but in actuality, no clear lines would be seen.

(54) Inside the body **10** is a coil **30**. The coil **30** is formed by multiple coil conductors **31** electrically coupled together and is, for example, solenoidal in shape. The coil conductors **31** are stacked in the length direction L together with the insulating layers **15**. It should be noted that in FIG. 7, the shape of the coil **30**, the positions of the coil conductors **31**, the coupling between the coil conductors **31**, etc., are not illustrated exactly. For example, coil conductors **31** adjacent in the length direction L are electrically coupled together by via conductors not illustrated in FIG. 7.

(55) The coil **30** has a coil axis C. The coil axis C of the coil **30** extends in the length direction L and runs between the first and second end surfaces **11a** and **11b** of the body **10** at the same time. That is, the direction of the coil axis C of the coil **30** is parallel with the first primary surface **12a**, which is the mounting surface, of the body **10**. The coil axis C of the coil **30**, furthermore, passes through the geometric center of the coil **30** when viewed in the length direction L.

(56) In this way, the direction of stacking of the insulating layers **15** and the direction of the coil axis C of the coil **30** are parallel with the first primary surface **12a**, which is the mounting surface, of the body **10**.

(57) The direction of stacking of the insulating layers **15** and the direction of the coil axis C of the coil **30** may be parallel with the length direction L as illustrated in FIG. 7 or may not. For example, it may be that the direction of stacking of the insulating layers **15** is parallel with the width direction W, and the direction of the coil axis C of the coil **30** is parallel with the length direction L. Even in this case, the direction of stacking of the insulating layers **15** and the direction of the coil axis C of the coil **30** are parallel with the first primary surface **12a**, which is the mounting surface, of the body **10**.

(58) The multilayer coil component **1** may further have a first interlinking conductor **41** and a second interlinking conductor **42**.

(59) The first interlinking conductor **41** is formed by multiple via conductors, not illustrated in FIG. 7, electrically coupled together and stacked in the length direction L together with insulating layers **15**. The first interlinking conductor **41** is exposed on the first end surface **11a** of the body **10**.

(60) The first outer electrode **21** is electrically coupled to the coil **30** by the first interlinking conductor **41**. Of the multiple coil conductors **31**, the closest to the first end surface **11a** of the body **10** is a coil conductor **31a**. That is, the first outer electrode **21** is electrically coupled to a coil conductor **31a** by the first interlinking conductor **41**.

(61) The first interlinking conductor **41** couples the first outer electrode **21** and the coil **30** together. Preferably, the first interlinking conductor **41** provides straight-line coupling between the first outer electrode **21** and the coil **30**, between the first outer electrode **21** and the coil conductor **31a** here. Preferably, when viewed in the length direction L, the first interlinking conductor **41** overlaps the coil conductor **31a** and is closer than the coil axis C to the first primary surface **12a**, which is the mounting surface, of the body **10**. These make electrical coupling between the first outer electrode **21** and the coil **30** easier.

(62) The first interlinking conductor **41** providing straight-line coupling between the first outer electrode **21** and the coil **30** means that the via conductors forming the first interlinking conductor

41 overlap when viewed in the length direction **L**. The via conductors forming the first interlinking conductor **41**, therefore, do not need to be arranged strictly in a straight line.

(63) Preferably, the first interlinking conductor **41** is coupled to the portion of the coil conductor **31a** closest to the first primary surface **12a** of the body **10**. This helps reduce the area of the first outer electrode **21** on the first end surface **11a** of the body **10**. The resulting decrease in the stray capacitance between the first outer electrode **21** and the coil **30** improves the radiofrequency characteristics of the multilayer coil component **1** accordingly.

(64) There may be only one first interlinking conductor **41**, or there may be multiple ones.

(65) The second interlinking conductor **42** is formed by multiple via conductors, not illustrated in FIG. 7, electrically coupled together and stacked in the length direction **L** together with insulating layers **15**. The second interlinking conductor **42** is exposed on the second end surface **11b** of the body **10**.

(66) The second outer electrode **22** is electrically coupled to the coil **30** by the second interlinking conductor **42**. Of the multiple coil conductors **31**, the closest to the second end surface **11b** of the body **10** is a coil conductor **31d**. That is, the second outer electrode **22** is electrically coupled to a coil conductor **31d** by the second interlinking conductor **42**.

(67) The second interlinking conductor **42** couples the second outer electrode **22** and the coil **30** together. Preferably, the second interlinking conductor **42** provides straight-line coupling between the second outer electrode **22** and the coil **30**, between the second outer electrode **22** and the coil conductor **31d** here. Preferably, when viewed in the length direction **L**, the second interlinking conductor **42** overlaps the coil conductor **31d** and is closer than the coil axis **C** to the first primary surface **12a**, which is the mounting surface, of the body **10**. These make electrical coupling between the second outer electrode **22** and the coil **30** easier.

(68) The second interlinking conductor **42** providing straight-line coupling between the second outer electrode **22** and the coil **30** means that the via conductors forming the second interlinking conductor **42** overlap when viewed in the length direction **L**. The via conductors forming the second interlinking conductor **42**, therefore, do not need to be arranged strictly in a straight line.

(69) Preferably, the second interlinking conductor **42** is coupled to the portion of the coil conductor **31d** closest to the first primary surface **12a** of the body **10**. This helps reduce the area of the second outer electrode **22** on the second end surface **11b** of the body **10**. The resulting decrease in the stray capacitance between the second outer electrode **22** and the coil **30** improves the radiofrequency characteristics of the multilayer coil component **1** accordingly.

(70) There may be only one second interlinking conductor **42**, or there may be multiple ones.

(71) Preferably, the dimension $L_{sub.3}$ of the coil **30** in the length direction **L** is 85% or more and 94% or less (i.e., from 85% to 94%), more preferably 90% or more and 94% or less (i.e., from 90% to 94%), of the dimension $L_{sub.2}$ of the body **10** in the length direction **L**.

(72) The dimension $L_{sub.3}$ of the coil **30** in the length direction **L** indicates the distance in the length direction **L** from the coil conductor **31a**, electrically coupled to the first outer electrode **21** by the first interlinking conductor **41**, to the coil conductor **31d**, electrically coupled to the second outer electrode **22** by the second interlinking conductor **42** (including the dimension of the coil conductors **31a** and **31d** in the length direction **L**). That is, the dimension $L_{sub.3}$ of the coil **30** in the length direction **L** indicates the dimension in the length direction **L** of the region over which the coil conductors **31** are present.

(73) If the dimension $L_{sub.3}$ of the coil **30** in the length direction **L** is smaller than 85% of the dimension $L_{sub.2}$ of the body **10** in the length direction **L**, a large stray capacitance of the coil **30** can affect the radiofrequency characteristics of the multilayer coil component **1**. If the dimension $L_{sub.3}$ of the coil **30** in the length direction **L** is larger than 94% of the dimension $L_{sub.2}$ of the body **10** in the length direction **L**, a large stray capacitance between the first outer electrode **21** and the coil **30** and that between the second outer electrode **22** and the coil **30** can affect the radiofrequency characteristics of the multilayer coil component **1**.

(74) FIG. 8 is a perspective schematic diagram illustrating an example of a disassembled form of the body and coil in FIG. 7. FIG. 9 is a plan schematic diagram illustrating an example of a disassembled form of the body and coil in FIG. 7.

(75) In the example illustrated in FIGS. 8 and 9, the body 10 is formed by insulating layers 15a, 15b, 15c, 15d, and 15e as the insulating layers 15 stacked in a direction of stacking, in the length direction L here.

(76) When what is stated applies to all of the insulating layers 15a, 15b, 15c, 15d, and 15e herein, the term insulating layers 15 is used.

(77) On the primary surface of the insulating layers 15a, 15b, 15c, and 15d are coil conductors 31a, 31b, 31c, and 31d, respectively, as the coil conductors 31. The coil conductors 31a, 31b, 31c, and 31d are stacked in the length direction L together with the insulating layers 15a, 15b, 15c, and 15d, and each coil conductor is electrically coupled to one another.

(78) When what is stated applies to all of the coil conductors 31a, 31b, 31c, and 31d herein, the term coil conductors 31 is used.

(79) In the example illustrated in FIGS. 8 and 9, the length of each of the coil conductors 31a, 31b, 31c, and 31d is the length of $\frac{3}{4}$ turns of the coil 30. That is, the number of coil conductors that need to be stacked to form three turns of the coil 30 is four. Inside the body 10, coil conductors 31a, 31b, 31c, and 31d are stacked repeatedly as one unit (equivalent of three turns).

(80) At both ends of the coil conductors 31, there may be a land portion. More specifically, there may be a land portion at both ends of the coil conductors 31a, 31b, 31c, and 31d.

(81) When viewed in the length direction L, the land portions of the coil conductors 31 may be round or may be polygonal.

(82) Insulating layers 15a, 15b, 15c, and 15d have via conductors 34a, 34b, 34c, and 34d, respectively, penetrating therethrough in the length direction L.

(83) The via conductors 34a, 34b, 34c, and 34d are coupled to one end of the coil conductors 31a, 31b, 31c, and 31d, respectively. If there is a land portion at both ends of the coil conductors 31a, 31b, 31c, and 31d as stated above, this means the via conductors 34a, 34b, 34c, and 34d are coupled to a land portion of the coil conductors 31a, 31b, 31c, and 31d, respectively.

(84) An insulating layer 15a, having a coil conductor 31a and a via conductor 34a, an insulating layer 15b, having a coil conductor 31b and a via conductor 34b, an insulating layer 15c, having a coil conductor 31c and a via conductor 34c, and an insulating layer 15d, having a coil conductor 31d and a via conductor 34d, are stacked repeatedly as one unit (a portion enclosed by dotted lines in FIGS. 8 and 9). This makes the coil conductors 31a, 31b, 31c, and 31d electrically coupled together by the via conductors 34a, 34b, 34c, and 34d. That is, coil conductors adjacent in the length direction L are electrically coupled together by via conductors.

(85) In this way, a solenoidal coil 30 disposed inside the body 10 is constructed.

(86) When viewed in the length direction L, the coil 30 may be round or may be polygonal. If the coil 30 includes land portions, for example if there is a land portion at both ends of the coil conductors 31, the shape of the coil 30 indicates the shape excluding the land portions.

(87) Preferably, when viewed in the length direction L, the inner diameter of the coil conductors 31 is 15% or more and 40% or less (i.e., from 15% to 40%) of the dimension W.sub.2 of the body 10 in the width direction W. The inner diameter of the coil conductors 31 is synonymous with the coil diameter of the coil 30. If the coil 30 is polygonal when viewed in the length direction L, the diameter of a circle equivalent in area to the polygon is deemed to be the coil diameter of the coil 30, or the inner diameter of the coil conductors 31.

(88) Preferably, the number of turns of the coil 30 is 35 or more, more preferably 35 or more and 45 or less (i.e., from 35 to 45). Thirty-five or more turns in the coil 30 give the multilayer coil component 1 a high impedance, and therefore a high transmission coefficient S21 in the radiofrequency band, thereby improving the radiofrequency characteristics of the multilayer coil component 1.

(89) An insulating layer **15e** has a via conductor **34e** penetrating therethrough in the length direction L.

(90) On the primary surface of the insulating layer **15e**, there may be a land portion coupled to the via conductor **34e**.

(91) Multiple ones of these insulating layers **15e**, having a via conductor **34e**, are stacked to cover the insulating layer **15a**, having a coil conductor **31a** and a via conductor **34a**, located at one end of the coil **30**. This makes the via conductors **34e** electrically coupled together to form the first interlinking conductor **41**, making the first interlinking conductor **41** exposed on the first end surface **11a** of the body **10**. As a result, the first outer electrode **21** and the coil conductor **31a** are electrically coupled together by the first interlinking conductor **41**.

(92) Multiple ones of these insulating layers **15e**, having a via conductor **34e**, are stacked to cover the insulating layer **15d**, having a coil conductor **31d** and a via conductor **34d**, located at the other end of the coil **30**. This makes the via conductors **34e** electrically coupled together to form the second interlinking conductor **42**, making the second interlinking conductor **42** exposed on the second end surface **11b** of the body **10**. As a result, the second outer electrode **22** and the coil conductor **31d** are electrically coupled together by the second interlinking conductor **42**.

(93) Preferably, the dimension of the first interlinking conductor **41** in the length direction L and that of the second interlinking conductor **42** are each 2.5% or more and 7.5% or less (i.e., from 2.5% to 7.5%), more preferably 2.5% or more and 5.0% or less (i.e., from 2.5% to 5.0%), of the dimension L.sub.2 of the body **10** in the length direction L. The resulting small inductance of the first and second interlinking conductors **41** and **42** improves the radiofrequency characteristics of the multilayer coil component **1**.

(94) Preferably, the dimension of the first interlinking conductor **41** in the width direction W and that of the second interlinking conductor **42** are each 8.0% or more and 20% or less (i.e., from 8.0% to 20%) of the dimension W.sub.2 of the body **10** in the width direction W.

(95) For when the multilayer coil component **1** is the 0603, 0402, or 1005 size, the following presents specific examples of preferred dimensions of the coil conductors **31**, first interlinking conductor **41**, and second interlinking conductor **42**.

(96) (1) When the multilayer coil component **1** is the 0603 size Preferably, when viewed in the length direction L, the inner diameter of the coil conductors **31** is 50 μm or more and 100 μm or less (i.e., from 50 μm to 100 μm). Preferably, the dimension of the first interlinking conductor **41** in the length direction L and that of the second interlinking conductor **42** are each 15 μm or more and 45 μm or less (i.e., from 15 μm to 45 μm), more preferably 15 μm or more and 30 μm or less (i.e., from 15 μm to 30 μm). Preferably, the dimension of the first interlinking conductor **41** in the width direction W and that of the second interlinking conductor **42** are each 30 μm or more and 60 μm or less (i.e., from 30 μm to 60 μm).

(97) (2) When the multilayer coil component **1** is the 0402 size Preferably, when viewed in the length direction L, the inner diameter of the coil conductors **31** is 30 μm or more and 70 μm or less (i.e., from 30 μm to 70 μm). Preferably, the dimension of the first interlinking conductor **41** in the length direction L and that of the second interlinking conductor **42** are each 10 μm or more and 30 μm or less (i.e., from 10 μm to 30 μm), more preferably 10 μm or more and 25 μm or less (i.e., from 10 μm to 25 μm). Preferably, the dimension of the first interlinking conductor **41** in the width direction W and that of the second interlinking conductor **42** are each 20 μm or more and 40 μm or less (i.e., from 20 μm to 40 μm).

(98) (3) When the multilayer coil component **1** is the 1005 size Preferably, when viewed in the length direction L, the inner diameter of the coil conductors **31** is 80 μm or more and 170 μm or less (i.e., from 80 μm to 170 μm). Preferably, the dimension of the first interlinking conductor **41** in the length direction L and that of the second interlinking conductor **42** are each 25 μm or more and 75 μm or less (i.e., from 25 μm to 75 μm), more preferably 25 μm or more and 50 μm or less (i.e., from 25 μm to 50 μm). Preferably, the dimension of the first interlinking conductor **41** in the width

direction W and that of the second interlinking conductor **42** are each 40 μm or more and 100 μm or less (i.e., from 40 μm to 100 μm).

(99) As for the material for forming the coil conductors **31a**, **31b**, **31c**, and **31d** and the via conductors **34a**, **34b**, **34c**, **34d**, and **34e**, examples include Ag, Au, Cu, Pd, Ni, Al, and alloys containing at least one of these metals.

(100) For the multilayer coil component according to the present disclosure, at least part of the body has a magnetic phase containing Fe, Ni, Zn, and Cu and a nonmagnetic phase containing Si.

(101) For the multilayer coil component **1**, at least part of the body **10** has a magnetic phase containing Fe, Ni, Zn, and Cu and a nonmagnetic phase containing Si.

(102) The magnetic phase may further contain Co, Bi, Sn, Mn, etc.

(103) Preferably, the magnetic phase is formed by a Ni—Cu—Zn ferrite material.

(104) Preferably, the Ni—Cu—Zn ferrite material contains, in a total of 100 mol %, 40 mol % or more and 49.5 mol % or less (i.e., from 40 mol % to 49.5 mol %) Fe in terms of an Fe.sub.2O.sub.3 basis, 10 mol % or more and 45 mol % or less (i.e., from 10 mol % to 45 mol %) Ni in terms of a NiO basis, 2 mol % or more and 35 mol % or less (i.e., from 2 mol % to 35 mol %) Zn in terms of a ZnO basis, and 6 mol % or more and 13 mol % or less (i.e., from 6 mol % to 13 mol %) Cu in terms of a CuO basis.

(105) The Ni—Cu—Zn ferrite material may further contain, for example, dopants or inevitable impurities, such as Co, Bi, Sn, and Mn.

(106) Preferably, the nonmagnetic phase is formed by a borosilicate glass material.

(107) Preferably, the borosilicate glass material contains, in a total of 100% by weight, 70% by weight or more and 85% by weight or less (i.e., from 70% by weight to 85% by weight) Si in terms of a SiO.sub.2 basis, 10% by weight or more and 25% by weight or less (i.e., from 10% by weight to 25% by weight) B in terms of a B.sub.2O.sub.3 basis, 0.5% by weight or more and 5% by weight or less (i.e., from 0.5% by weight to 5% by weight) alkali metal A in terms of an A.sub.2O basis, and 0% by weight or more and 5% by weight or less (i.e., from 0% by weight to 5% by weight) Al in terms of an Al.sub.2O.sub.3 basis.

(108) The borosilicate glass material may further contain forsterite ($2\text{MgO.Math.SiO.sub.2}$), quartz (SiO.sub.2), etc., as filler.

(109) The nonmagnetic phase may be formed by an oxide represented by $a\text{ZnO.Math.SiO.sub.2}$ (a is 1.8 or more and 2.2 or less (i.e., from 1.8 to 2.2)). Examples of such oxides include the Zn.sub.2SiO.sub.4 called willemite. Such an oxide may contain Cu in place of part of Zn.

(110) The magnetic and nonmagnetic phases are differentiated as follows. First, the multilayer coil component is polished to substantially the middle in the width direction to expose a cross-section parallel with the length and height directions like that illustrated in, for example, FIG. 7. Then the exposed cross-section of the body is subjected to elemental mapping by scanning transmission electron microscopy-energy-dispersive X-ray spectroscopy (STEM-EDX). Then the two phases are differentiated: the regions in which Fe is present constitute the magnetic phase, and the regions in which Si is present constitute the nonmagnetic phase.

(111) FIG. 10 is a side schematic view of the multilayer coil component in FIG. 1.

(112) In the example illustrated in FIG. 10, when, for the multilayer coil component **1**, a boundary G is defined that extends parallel with the height and width directions T and W at the midpoint of the body **10** in the length direction L, the body **10** has first and second body sections **61** and **62** aligned in the length direction L with the boundary G therebetween, the first body section **61** including the first end surface **11a** and the second body section **62** including the second end surface **11b**. That is, the body **10** is divided into two, first and second body sections **61** and **62**, by a boundary G in the length direction L.

(113) Regarding the boundary G, the midpoint of the body **10** in the length direction L is determined as the point that bisects the maximum dimension of the body **10** in the length direction L.

(114) The first outer electrode **21** is on the surface of the first body section **61**.

(115) The second outer electrode **22** is on the surface of the second body section **62**.

(116) For the multilayer coil component according to the present disclosure, the first outer electrode has, in order from the body side, an underlying electrode and a plating electrode on the underlying electrode, and the end in the length direction of the portion of the plating electrode lying on the first primary surface of the body is closer to the second end surface of the body than the end in the length direction of the portion of the underlying electrode lying on the first primary surface of the body.

(117) FIG. **11** is a schematic diagram illustrating part of a cross-section of the first body section and first outer electrode in FIG. **10** parallel with the length and height directions.

(118) As illustrated in FIG. **11**, the first outer electrode **21** has, in order from the body **10** side, an underlying electrode **21a** and a plating electrode **21b** on the underlying electrode **21a**.

(119) The end **21ba** in the length direction L of the portion of the plating electrode **21b** lying on the first primary surface **12a** of the body **10** is closer to the second end surface **11b** of the body **10** (right in FIG. **11**) than the end **21aa** in the length direction L of the portion of the underlying electrode **21a** lying on the first primary surface **12a** of the body **10**.

(120) The end in the length direction of the underlying electrode lying on the first primary surface of the body and the end in the length direction of the plating electrode lying on the first primary surface of the body are determined on a cross-section of the multilayer coil component extending parallel with the length and height directions substantially in the middle in the width direction.

(121) A plating electrode in an outer electrode having underlying and plating electrodes like those of the multilayer coil component according to the present disclosure can stretch much larger than the underlying electrode along the surface of the body. If an outer electrode is formed with such an overgrown plating electrode, the stray capacity between the outer electrode and the coil can be so large that it will affect the radiofrequency characteristics of the multilayer coil component.

(122) To address this, for the multilayer coil component according to the present disclosure, the distance in the length direction between the end of the plating electrode and the end of the underlying electrode is 30 μm or less.

(123) For the multilayer coil component **1**, the distance a in the length direction L between the end **21ba** in the length direction L of the portion of the plating electrode **21b** lying on the first primary surface **12a** of the body **10** and the end **21aa** in the length direction L of the portion of the underlying electrode **21a** lying on the first primary surface **12a** of the body **10** is 30 μm or less. More specifically, the distance a in the length direction L between the end **21ba** of the plating electrode **21b** and the end **21aa** of the underlying electrode **21a** is greater than 0 μm and 30 μm or less (i.e., from 0 μm to 30 μm). This means the plating electrode **21b** does not stretch much larger than the underlying electrode **21a** along the first primary surface **12a** of the body **10**. The stray capacity between the coil **30** and the first outer electrode **21**, therefore, is prevented from being large; as a result, the radiofrequency characteristics of the multilayer coil component **1** are improved accordingly.

(124) If the distance a in the length direction L between the end **21ba** of the plating electrode **21b** and the end **21aa** of the underlying electrode **21a** is greater than 30 μm , the plating electrode **21b** stretches much larger than the underlying electrode **21a** along the first primary surface **12a** of the body **10**, and the stray capacity between the coil **30** and the first outer electrode **21** is not prevented from being large.

(125) Overall, the multilayer coil component according to the present disclosure has an outer electrode, a first outer electrode here, having a plating electrode but at the same time is configured such that the distance in the length direction between the end of the plating electrode and the end of the underlying electrode is 30 μm or less. By virtue of this, the multilayer coil component is superior in radiofrequency characteristics. That is, the multilayer coil component according to the present disclosure has a high transmission coefficient S21 in the radiofrequency band and is

consistent in the transmission coefficient S21 in the radiofrequency and lower-frequency bands.

(126) Preferably, the underlying electrode **21a** contains Ag.

(127) Preferably, the plating electrode **21b** contains at least one of Ni or Sn.

(128) The plating electrode **21b** may have a single-layer structure or may have a multilayer structure, but preferably has a multilayer structure. If the plating electrode **21b** has a multilayer structure, it is preferred that the plating electrode **21b** have, in order from the underlying electrode **21a** side, a Ni plating electrode and a Sn plating electrode.

(129) Plating electrodes like the plating electrode **21b** are differentiated from underlying electrodes formed by non-plating processes like the underlying electrode **21a** by elemental analysis by energy-dispersive X-ray spectroscopy (EDX) and based on information such as being dense in a cross-section parallel with the length and height directions like that illustrated in FIG. **11**.

(130) Preferably, when a first reference position **H1** is defined that coincides in the height direction **T** with the end **21aa** of the underlying electrode **21a**, the first body section **61** has a first region **61a** including at least the range having a dimension $P_{sub.1}$ in the length direction **L** of 20 μm from the first reference position **H1** toward the second end surface **11b** (to the right in FIG. **11**). That is, it is preferred that the first region **61a** include at least the range from the point at which the distance $P_{sub.1}$ in the length direction **L** from the first reference position **H1** toward the second end surface **11b** is 0 μm to the point at which the distance $P_{sub.1}$ is 20 μm .

(131) Preferably, the end **61aa** of the first region **61a** on the second end surface **11b** side is at the point where the distance $P_{sub.1}$ in the length direction **L** from the first reference position **H1** is 20 μm or more and 60 μm or less (i.e., from 20 μm to 60 μm). That is, it is preferred that the first region **61a** include a range from the point at which the distance $P_{sub.1}$ in the length direction **L** from the first reference position **H1** toward the second end surface **11b** is 0 μm to the point at which the distance $P_{sub.1}$ is $\alpha_{sub.1} \mu\text{m}$ ($20 \leq \alpha_{sub.1} \leq 60$).

(132) Preferably, the first region **61a** further includes at least the range having a dimension $P_{sub.2}$ in the length direction **L** of 20 μm from the first reference position **H1** toward the first end surface **11a** (to the left in FIG. **11**). That is, it is preferred that the first region **61a** further include at least the range from the point at which the distance $P_{sub.2}$ in the length direction **L** from the first reference position **H1** toward the first end surface **11a** is 0 μm to the point at which the distance $P_{sub.2}$ is 20 μm .

(133) Preferably, the end **61ab** of the first region **61a** on the first end surface **11a** side is at the point where the distance $P_{sub.2}$ in the length direction **L** from the first reference position **H1** is 20 μm or more and 60 μm or less (i.e., from 20 μm to 60 μm). That is, it is preferred that the first region **61a** further include a range from the point at which the distance $P_{sub.2}$ in the length direction **L** from the first reference position **H1** toward the first end surface **11a** is 0 μm to the point at which the distance $P_{sub.2}$ is $\alpha_{sub.2} \mu\text{m}$ ($20 \leq \alpha_{sub.2} \leq 60$).

(134) Preferably, the first region **61a** is totally the nonmagnetic phase or has both the nonmagnetic and magnetic phases. This prevents the plating electrode **21b** from greatly outgrowing the underlying electrode **21a** along the first primary surface **12a** of the body **10** during its formation, making it more certain that the distance a in the length direction **L** between the end **21ba** of the plating electrode **21b** and the end **21aa** of the underlying electrode **21a** is 30 μm or less. The end **21ba** of the plating electrode **21b**, therefore, is forced to stay, preferably at a point where it is within the first region **61a** as illustrated in FIG. **11**.

(135) Preferably, in the first region **61a**, the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases is 60% by volume or more. More specifically, it is preferred that the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases in the first region **61a** be 60% by volume or more and 100% by volume or less (i.e., from 60% by volume to 100% by volume). This makes it more certain that the distance a in the length direction **L** between the end **21ba** of the plating electrode **21b** and the end **21aa** of the underlying electrode **21a** is 30 μm or less when the plating electrode

21b is formed.

(136) Preferably, the first region **61a** contains, in a total of 100% by weight, 30.0% by weight or more Si in terms of a SiO₂ basis. This makes it more certain that the distance *a* in the length direction *L* between the end **21ba** of the plating electrode **21b** and the end **21aa** of the underlying electrode **21a** is 30 μm or less when the plating electrode **21b** is formed.

(137) Preferably, the first region **61a** contains, in a total of 100% by weight, 85.0% by weight or less Si in terms of a SiO₂ basis.

(138) Preferably, the first region **61a** contains, in a total of 100% by weight, 30.0% by weight or more and 85.0% by weight or less (i.e., from 30.0% by weight to 85.0% by weight) Si in terms of a SiO₂ basis, 4.0% by weight or more and 15.0% by weight or less (i.e., from 4.0% by weight to 15.0% by weight) B in terms of a B₂O₃ basis, 0% by weight or more and 45.0% by weight or less (i.e., from 0% by weight to 45.0% by weight) Fe in terms of an Fe₂O₃ basis, 0% by weight or more and 15.0% by weight (i.e., from 0% by weight to 15.0% by weight) Ni in terms of a NiO basis, 0% by weight or more and 8.0% by weight or less (i.e., from 0% by weight to 8.0% by weight) Zn in terms of a ZnO basis, and 0% by weight or more and 5.0% by weight or less (i.e., from 0% by weight to 5.0% by weight) Cu in terms of a CuO basis.

(139) Preferably, the first region **61a** further contains, in a total of 100% by weight, 0.3% by weight or more and 1.5% by weight or less (i.e., from 0.3% by weight to 1.5% by weight) K in terms of a K₂O basis and 0.9% by weight or more and 3.5% by weight or less (i.e., from 0.9% by weight to 3.5% by weight) Mg in terms of a MgO basis.

(140) Preferably, the first body section **61** has a second region **61b**, which is all but the first region **61a**.

(141) Preferably, the second region **61b** is totally the magnetic phase or has both the magnetic and nonmagnetic phases. In that case the impedance of the multilayer coil component **1** tends to be high. As a result, the radiofrequency characteristics of the multilayer coil component **1** tend to be improved.

(142) Preferably, in the second region **61b**, the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases is 50% by volume or less. More specifically, it is preferred that the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases in the second region **61b** be 0% by volume or more and 50% by volume or less (i.e., from 0% by volume to 50% by volume). In that case the percentage by volume of the magnetic phase to the total volume of the magnetic and nonmagnetic phase in the second region **61b** is at least comparable to the percentage by volume of the nonmagnetic phase there, and, therefore, the impedance of the multilayer coil component **1** tends to be high. As a result, the radiofrequency characteristics of the multilayer coil component **1** tend to be improved.

(143) In each of the first and second regions, the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases is determined as follows. First, the multilayer coil component is polished to substantially the middle in the width direction to expose a cross-section parallel with the length and height directions like that illustrated in, for example, FIGS. 7 and 11. Then, if the percentage by volume of the nonmagnetic phase in the first region is measured on the exposed cross-section in the first body section, the range having a dimension in the length direction of 20 μm from the first reference position toward the second end surface is selected as the region of interest. If the percentage by volume of the nonmagnetic phase in the second region is measured on the exposed cross-section in the first body section, the range having a dimension in the length direction of 20 μm from the first end surface toward the second end surface is selected as the region of interest. Within that region of interest three 20-μm square areas are picked, and then the magnetic and nonmagnetic phases are differentiated as described above by performing elemental mapping by scanning transmission electron microscopy-energy-dispersive X-ray spectroscopy. After that, in each of these three areas, the percentage by area of the nonmagnetic

phase to the total area of the magnetic and nonmagnetic phases is measured on the elemental mapping image using image analysis software. Then these measured percentages by area are averaged, and the calculated average is reported as the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases.

(144) Preferably, the second region **61b** contains, in a total of 100% by weight, 25.0% by weight or less Si in terms of a SiO_2 basis. In that case the impedance of the multilayer coil component **1** tends to be high. As a result, the radiofrequency characteristics of the multilayer coil component **1** tend to be improved.

(145) It should be noted that the second region **61b** may be Si-free.

(146) Preferably, the second region **61b** contains, in a total of 100% by weight, 0% by weight or more and 25.0% by weight or less (i.e., from 0% by weight to 25.0% by weight) Si in terms of a SiO_2 basis, 0% by weight or more and 5.0% by weight or less (i.e., from 0% by weight to 5.0% by weight) B in terms of a B_2O_3 basis, 45.0% by weight or more and 70.0% by weight or less (i.e., from 45.0% by weight to 70.0% by weight) Fe in terms of an Fe_2O_3 basis, 10.0% by weight or more and 20.0% by weight or less (i.e., from 10.0% by weight to 20.0% by weight) Ni in terms of a NiO basis, and 5.0% by weight or more and 12.0% by weight or less (i.e., from 5.0% by weight to 12.0% by weight) Zn in terms of a ZnO basis.

(147) Preferably, the Si content of the first region **61a** in a total of 100% by weight is larger than the Si content of the second region **61b** in a total of 100% by weight by 7.0% by weight or more in terms of a SiO_2 basis. That is, it is preferred that the difference between the Si content of the first region **61a** in a total of 100% by weight and the Si content of the second region **61b** in a total of 100% by weight be 7.0% by weight or more in terms of a SiO_2 basis. Preferably, furthermore, the difference between the Si content of the first region **61a** in a total of 100% by weight and the Si content of the second region **61b** in a total of 100% by weight is 60.0% by weight or less. The Si content of the second region **61b** in a total of 100% by weight may be 0% by weight.

(148) The composition of the first and second regions is checked by analyzing each region of interest described above by inductively coupled plasma atomic emission spectroscopy/mass spectrometry (ICP-AES/MS).

(149) The first body section **61** may be totally the first region **61a**, although it is preferred that it have the first and second regions **61a** and **61b**.

(150) Preferably, the second body section **62** and the second outer electrode **22**, too, have the same structure and construction as the first body section **61** and the first outer electrode **21** as presented below.

(151) FIG. **12** is a schematic diagram illustrating part of a cross-section of the second body section and second outer electrode in FIG. **10** parallel with the length and height directions.

(152) As illustrated in FIG. **12**, the second outer electrode **22** has, in order from the body **10** side, an underlying electrode **22a** and a plating electrode **22b** on the underlying electrode **22a**.

(153) The end **22ba** in the length direction L of the portion of the plating electrode **22b** lying on the first primary surface **12a** of the body **10** is closer to the first end surface **11a** of the body **10** (left in FIG. **11**) than the end **22aa** in the length direction L of the portion of the underlying electrode **22a** lying on the first primary surface **12a** of the body **10**.

(154) The distance b in the length direction L between the end **22ba** in the length direction L of the portion of the plating electrode **22b** lying on the first primary surface **12a** of the body **10** and the end **22aa** in the length direction L of the portion of the underlying electrode **22a** lying on the first primary surface **12a** of the body **10** is 30 μm or less. More specifically, the distance b in the length direction L between the end **22ba** of the plating electrode **22b** and the end **22aa** of the underlying electrode **22a** be greater than 0 μm and 30 μm or less (i.e., from 0 μm to 30 μm).

(155) A preferred structure and construction for the underlying and plating electrodes **22a** and **22b** is the same as a preferred structure and construction for the underlying and plating electrodes **21a** and **21b**, respectively.

- (156) Preferably, when a second reference position H2 is defined that coincides in the height direction T with the end **22aa** of the underlying electrode **22a**, the second body section **62** has a first region **62a** including at least the range having a dimension Q.sub.1 in the length direction L of 20 μm from the second reference position H2 toward the first end surface **11a** (to the left in FIG. 12). That is, it is preferred that the first region **62a** include at least the range from the point at which the distance Q.sub.1 in the length direction L from the second reference position H2 toward the first end surface **11a** is 0 μm to the point at which the distance Q.sub.1 is 20 μm .
- (157) Preferably, the end **62aa** of the first region **62a** on the first end surface **11a** side is at the point where the distance Q.sub.1 in the length direction L from the second reference position H2 is 20 μm or more and 60 μm or less (i.e., from 20 μm to 60 μm). That is, it is preferred that the first region **62a** include a range from the point at which the distance Q.sub.1 in the length direction L from the second reference position H2 toward the first end surface **11a** is 0 μm to the point at which the distance Q.sub.1 is $\beta_{\text{sub.1}}$ μm ($20 \leq \beta_{\text{sub.1}} \leq 60$).
- (158) Preferably, the first region **62a** further includes at least the range having a dimension Q.sub.2 in the length direction L of 20 μm from the second reference position H2 toward the second end surface **11b** (to the right in FIG. 12). That is, it is preferred that the first region **62a** further include at least the range from the point at which the distance Q.sub.2 in the length direction L from the second reference position H2 toward the second end surface **11b** is 0 μm to the point at which the distance Q.sub.2 is 20 μm .
- (159) Preferably, the end **62ab** of the first region **62a** on the second end surface **11b** side is at the point where the distance Q.sub.2 in the length direction L from the second reference position H2 is 20 μm or more and 60 μm or less (i.e., from 20 μm to 60 μm). That is, it is preferred that the first region **62a** further include a range from the point at which the distance Q.sub.2 in the length direction L from the second reference position H2 toward the second end surface **11b** is 0 μm to the point at which the distance Q.sub.2 is $\beta_{\text{sub.2}}$ μm ($20 \leq \beta_{\text{sub.2}} \leq 60$).
- (160) Preferably, the first region **62a** is totally a nonmagnetic phase or has both nonmagnetic and magnetic phases.
- (161) Preferably, the second body section **62** has a second region **62b**, which is all but the first region **62a**.
- (162) Preferably, the second region **62b** is totally a magnetic phase or has both magnetic and nonmagnetic phases.
- (163) A preferred percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases in each of the first and second regions **62a** and **62b** is the same as a preferred percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases in each of the first and second regions **61a** and **61b**.
- (164) For the second body section, too, the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases in each of the first and second regions is determined as described above. This time, if the percentage by volume of the nonmagnetic phase in the first region is measured on the exposed cross-section in the second body section, the range having a dimension in the length direction of 20 μm from the second reference position toward the first end surface is selected as the region of interest. If the percentage by volume of the nonmagnetic phase in the second region is measured on the exposed cross-section in the second body section, the range having a dimension in the length direction of 20 μm from the second end surface toward the first end surface is selected as the region of interest.
- (165) A preferred composition for the first and second regions **62a** and **62b** is the same as a preferred composition for the first and second regions **61a** and **61b**, respectively.
- (166) The second body section **62** may be totally the first region **62a**, although it is preferred that it have the first and second regions **62a** and **62b**.
- (167) The multilayer coil component **1** is produced by, for example, the following method.
- (168) Magnetic Material Preparation Step

(169) First, predetermined proportions of Fe.sub.2O.sub.3, NiO, ZnO, and CuO are weighed out.
(170) Then these measured materials are wet-mixed and then ground to give slurry. The duration of mixing of the materials is, for example, 4 hours or more and 8 hours or less (i.e., from 4 hours to 8 hours).

(171) Then the resulting slurry is dried and then calcined. The calcination temperature is, for example, 700° C. or above and 800° C. or below (i.e., from 700° C. to 800° C.). The duration of calcination is, for example, 2 hours or more and 5 hours or less (i.e., from 2 hours to 5 hours).

(172) In such a way, a powdery magnetic material, more specifically a powdery ferrite material, is prepared.

(173) Preferably, the ferrite material contains, in a total of 100 mol %, 40 mol % or more and 49.5 mol % or less (i.e., from 40 mol % to 49.5 mol %) Fe in terms of an Fe.sub.2O.sub.3 basis, 10 mol % or more and 45 mol % or less (i.e., from 10 mol % to 45 mol %) Ni in terms of a NiO basis, 2 mol % or more and 35 mol % or less (i.e., from 2 mol % to 35 mol %) Zn in terms of a ZnO basis, and 6 mol % or more and 13 mol % or less (i.e., from 6 mol % to 13 mol %) Cu in terms of a CuO basis.

(174) Nonmagnetic Material Preparation Step

(175) First, a borosilicate glass powder containing Si, B, an alkali metal, and Al in predetermined proportions is prepared.

(176) Preferably, the borosilicate glass material contains, in a total of 100% by weight, 70% by weight or more and 85% by weight or less (i.e., from 70% by weight to 85% by weight) Si in terms of a SiO.sub.2 basis, 10% by weight or more and 25% by weight or less (i.e., from 10% by weight to 25% by weight) B in terms of a B.sub.2O.sub.3 basis, 0.5% by weight or more and 5% by weight or less (i.e., from 0.5% by weight to 5% by weight) alkali metal A in terms of an A.sub.2O basis, and 0% by weight or more and 5% by weight or less (i.e., from 0% by weight to 5% by weight) Al in terms of an Al.sub.2O.sub.3 basis.

(177) Then, as filler, a forsterite powder and a quartz powder are prepared.

(178) Then the borosilicate glass, forsterite, and quartz powders are wet-mixed in predetermined proportions and then ground to give a nonmagnetic material.

(179) Green Sheet Preparation Step

(180) First, predetermined proportions of the magnetic and nonmagnetic materials are weighed out. Then these measured materials, an organic binder, such as a polyvinyl butyral resin, organic solvents, such as ethanol and toluene, a plasticizer, etc., are mixed together and then ground to give slurry. Then the resulting slurry is shaped, for example by doctor blading, into sheets having a predetermined thickness, and then a predetermined shape is punched out of them to give green sheets.

(181) In preparing the green sheets, the proportions of blending of the magnetic and nonmagnetic materials are adjusted so that type 1 green sheets, in which the percentage by volume of the magnetic material to the total volume of the magnetic and nonmagnetic materials is 60% by volume or more, and type 2 green sheets, in which the percentage by volume of the magnetic material to the total volume of the magnetic and nonmagnetic materials is 50% by volume or less, will result.

(182) In addition, in preparing the green sheets, the proportions of blending of the magnetic and nonmagnetic materials may be adjusted so that type 3 green sheets, which contain, in a total of 100% by weight, 30.0% by weight or more Si in terms of a SiO.sub.2 basis, and type 4 green sheets, which contain, in a total of 100% by weight, 25.0% by weight or less Si in terms of a SiO.sub.2 basis, will result.

(183) When what is stated applies to all of type 1, type 2, type 3, and type 4 green sheets hereinafter, the simple term "green sheet" is used.

(184) In the following steps, a case in which the green sheets are a combination of type 1 and type 2 green sheets is described. The same applies if the green sheets are a combination of type 3 and

type 4 green sheets.

(185) Conductor Pattern Formation Step

(186) First, a predetermined point of the green sheet is irradiated with a laser to create a via hole.

(187) Then an electrically conductive paste, such as Ag paste, is coated onto the surface of the green sheet and put into the via hole at the same time, for example by screen printing. Through this, a conductor pattern for via conductors is formed in the via hole in the green sheet, and a conductor pattern for coil conductors, coupled to the conductor pattern for via conductors, is formed on the surface of the green sheet at the same time. In such a way, a coil sheet is prepared as a green sheet with a conductor pattern for coil conductors formed thereon and a conductor pattern for via conductors formed therein. Multiple ones of these coil sheets are prepared, with each coil sheet having a conductor pattern for coil conductors corresponding to a coil conductor illustrated in FIGS. 8 and 9 formed thereon and a conductor pattern for via conductors corresponding to a via conductor illustrated in FIGS. 8 and 9 formed therein.

(188) Separately from the coil sheets, a via sheet is prepared as a green sheet with a conductor pattern for via conductors formed therein by putting an electrically conductive paste, such as Ag paste, into the via hole, for example by screen printing. Multiple ones of these via sheets, too, are prepared, with each via sheet having a conductor pattern for via conductors corresponding to a via conductor illustrated in FIGS. 8 and 9 formed therein.

(189) In preparing the coil sheets and via sheets, those that will be placed in the region the manufacturer wants to be the first region in the first and second body sections of the finished bodies are made using type 1 green sheets, and those that will be placed in the region the manufacturer wants to be the second region are made using type 2 green sheets.

(190) Multilayer Block Preparation Step

(191) The coil sheets and via sheets are stacked in the direction of stacking in an order corresponding to FIGS. 8 and 9 and then joined together by heat and pressure bonding to give a multilayer block. This makes the multilayer block have type 1 green sheets in the region the manufacturer wants to be the first region in the first and second body sections of the finished bodies and type 2 green sheets in the region the manufacturer wants to be the second region.

(192) Body and Coil Production Step

(193) First, the multilayer block is cut, for example with a dicer, into a predetermined size to give singulated chips.

(194) Then the singulated chips are fired. The firing temperature is, for example, 900° C. or above and 920° C. or below (i.e., from 900° C. to 920° C.). The duration of firing, is, for example, 2 hours or more and 4 hours or less (i.e., from 2 hours to 4 hours).

(195) Firing the singulated chips will turn the green sheets, which are coil sheets and via sheets, into insulating layers, thereby giving bodies that are formed by multiple insulating layers stacked in a direction of stacking, in the length direction here.

(196) As mentioned above, in preparing the multilayer block, type 1 green sheets are placed in the region the manufacturer wants to be the first region in the first and second body sections of the finished bodies, and type 2 green sheets are placed in the region the manufacturer wants to be the second region. In the first and second body sections of the bodies produced in this step, therefore, the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases in the first region is 60% by volume or more. In the second region, furthermore, the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases is 50% by volume or less.

(197) Firing the singulated chips will turn the conductor pattern for coil conductors on and the conductor pattern for via conductors in the coil sheets into a coil conductor and a via conductor, respectively, thereby giving coils that are formed by multiple coil conductors stacked in the length direction and electrically coupled together by via conductors.

(198) In this way, bodies and coils disposed inside the bodies are produced. The direction of

stacking of the insulating layers and the direction of the coil axis of the coil are parallel with the first primary surface, which is the mounting surface, of the body, parallel with the length direction here.

(199) Firing the singulated chips will turn the conductor pattern for via conductors in the via sheets into via conductors, thereby producing first and second interlinking conductors that are formed by multiple via conductors stacked in the length direction and electrically coupled together. The first interlinking conductor will become exposed on the first end surface of the body. The second interlinking conductor will become exposed on the second end surface of the body.

(200) The corners and edges of the bodies may be rounded, for example by barrel polishing.

(201) Outer Electrode Formation Step

(202) First, an electrically conductive paste containing Ag and glass frit is stretched to a predetermined thickness, and the bodies are dipped diagonally into the resulting layer. Then the resulting coating is baked to form an underlying electrode that extends from part of the first end surface of the body to part of each of the first primary surface, first lateral surface, and second lateral surface. Likewise, an underlying electrode is formed that extends from part of the second end surface of the body to part of each of the first primary surface, first lateral surface, and second lateral surface. The temperature for baking the coating is, for example, 800° C. or above and 820° C. or below (i.e., from 800° C. to 820° C.).

(203) Then, for example by electrolytic plating, a Ni plating electrode and a Sn plating electrode are formed on each underlying electrode in sequence.

(204) During this, the end in the length direction of the portion of the plating electrode lying on the first primary surface of the plating electrode comes closer to the second end surface of the body than the end in the length direction of the portion of the underlying electrode lying on the first primary surface of the body. This is done in such a manner that the distance in the length direction between the end of the plating electrode, the end of the Sn plating electrode here, and the end of the underlying electrode will be 30 μm or less.

(205) An example of how to make the distance in the length direction between the end of the plating electrode and the end of the underlying electrode 30 μm or less is to place type 1 green sheets in the region the manufacturer wants to be the first region in the first and second body sections of the body in preparing the multilayer block. Using such a method ensures the percentage by volume of the magnetic phase to the total volume of the magnetic and nonmagnetic phases in the first region will be 60% by volume or more in the first and second body sections of the finished bodies, thereby making it more certain that the distance in the length direction between the end of the plating electrode and the end of the underlying electrode will be 30 μm or less when the plating electrode is formed.

(206) In such a way, a first outer electrode, electrically coupled to the coil by the first interlinking conductor, and a second outer electrode, electrically coupled to the coil by the second interlinking conductor, are formed.

(207) Through this, multilayer coil components 1 are produced.

Examples

(208) The following presents examples in which the multilayer coil component according to the present disclosure is disclosed more specifically. It should be noted that the present disclosure is not limited to these examples.

(209) Multilayer coil component samples 1 to 5 were produced by the following method.

(210) Magnetic Material Preparation Step

(211) First, proportions of Fe₂O₃, NiO, ZnO, and CuO were weighed out so that Fe₂O₃ would constitute 48.0 mol %, NiO 14.0 mol %, ZnO 30.0 mol %, and CuO 8.0 mol %. Then these measured materials were mixed in a ball mill with purified water and a PSZ (partially stabilized zirconia) medium for 6 hours and then ground to give slurry. Then the resulting slurry was dried and then calcined at 800° C. for 2 hours. In such a way, a powdery magnetic

material, more specifically a powdery ferrite material, was prepared.

(212) Nonmagnetic Material Preparation Step

(213) First, a borosilicate glass powder containing Si, B, an alkali metal, and Al in predetermined proportions was prepared. Then, as filler, a forsterite powder and a quartz powder were prepared. Then the borosilicate glass powder, forsterite powder, and quartz powder are wet-mixed in such proportions that the glass powder would constitute 72% by weight, forsterite 4% by weight, and quartz 24% by weight, and then ground to give a nonmagnetic material.

(214) Separately, body samples were prepared by the following method for checking the composition of the body of finished multilayer coil component samples 1 to 5 with. First, amounts of the magnetic and nonmagnetic materials were weighed out so that the magnetic material would constitute 100% by volume, and the nonmagnetic material 0% by volume. Then these measured materials, a polyvinyl butyral resin as an organic binder, ethanol and toluene as organic solvents, and a plasticizer were mixed together in a ball mill with a PSZ medium and then ground to give slurry. Then the resulting slurry was shaped into sheets by doctor blading, and these sheets were stacked and joined together by heat and pressure bonding to give a green block. A portion was punched out of the green block and fired at 910° C. for 4 hours, giving disk-shaped body sample A having a thickness of 0.5 mm and a diameter of 10 mm.

(215) Then body samples B, C, and D were prepared in the same way as body sample A, except that the magnetic and nonmagnetic materials were blended in the proportions specified in Table 1.

(216) Body samples A, B, C, and D were analyzed by inductively coupled plasma atomic emission spectroscopy/mass spectrometry, confirming that they had the compositions presented in Table 1.

(217) When looking at the compositions presented in Table 1, one can say that green sheets C and D are categorized into type 3 green sheets, which contain, in a total of 100% by weight, 30.0% by weight or more Si in terms of a SiO₂ basis. One can also say that green sheets A and B are categorized into type 4 green sheets, which contain, in a total of 100% by weight, 25.0% by weight or less Si in terms of a SiO₂ basis.

(218) TABLE-US-00001 TABLE 1 Body Samples A B C D Percentage by Magnetic material 100 50 40 0 volume Nonmagnetic 0 50 60 100 (% by volume) material Composition K₂O — 0.4 0.5 1.2 (% by weight) B₂O₃ — 3.8 4.9 12.9 SiO₂ — 23.9 31.4 82.3 Al₂O₃ — 0.1 0.1 0.2 MgO — 1.0 1.3 3.4 Fe₂O₃ 66.1 46.8 40.8 — NiO 18.8 13.3 11.6 — ZnO 9.7 6.9 6.0 — CuO 5.4 3.8 3.4 —

Green Sheet Preparation Step

(219) Green sheets A as building units for a green block as described above were prepared using materials having the same composition as body sample A. That is, in the finished bodies, a region formed by green sheets A would have the same composition as body sample A.

(220) Likewise, green sheets B were prepared using materials having the same composition as body sample B, green sheets C were prepared using materials having the same composition as body sample C, and green sheets D were prepared using materials having the same composition as body sample D. In the finished bodies, a region formed by green sheets B would have the same composition as body sample B, a region formed by green sheets C would have the same composition as body sample C, and a region formed by green sheets D would have the same composition as body sample D.

(221) Green sheets C and D are categorized into type 1 green sheets, in which the percentage by volume of the nonmagnetic material to the total volume of the magnetic and nonmagnetic materials is 60% by volume or more. Green sheets A and B are categorized into type 2 green sheets, in which the percentage by volume of the nonmagnetic material to the total volume of the magnetic and nonmagnetic materials is 50% by volume or less.

(222) Conductor Pattern Formation Step

(223) First, a predetermined point of the green sheet was irradiated with a laser to create a via hole.

(224) Then Ag paste was coated onto the surface of the green sheet and put into the via hole at the

same time by screen printing. Through this, a conductor pattern for via conductors was formed in the via hole in the green sheet, and a conductor pattern for coil conductors, coupled to the conductor pattern for via conductors, was formed on the surface of the green sheet at the same time. In such a way, a coil sheet was prepared as a green sheet with a conductor pattern for coil conductors formed thereon and a conductor pattern for via conductors formed therein. Multiple ones of these coil sheets were prepared, with each coil sheet given a conductor pattern for coil conductors corresponding to a coil conductor illustrated in FIGS. 8 and 9 formed thereon and a conductor pattern for via conductors corresponding to a via conductor illustrated in FIGS. 8 and 9 formed therein.

(225) Separately from the coil sheets, a via sheet was prepared as a green sheet with a conductor pattern for via conductors formed therein by putting Ag paste into the via hole by screen printing. Multiple ones of these via sheets, too, were prepared, with each via sheet given a conductor pattern for via conductors corresponding to a via conductor illustrated in FIGS. 8 and 9 formed therein.

(226) Multilayer Block Preparation Step

(227) The coil sheets and via sheets were stacked in the direction of stacking in an order corresponding to FIGS. 8 and 9 and then joined together by heat and pressure bonding to give a multilayer block.

(228) Body and Coil Production Step

(229) First, the multilayer block was cut with a dicer into a predetermined size to give singulated chips.

(230) Then the singulated chips were fired at 910° C. for 4 hours.

(231) Firing the singulated chips turned the green sheets, which were coil sheets and via sheets, into insulating layers, thereby giving bodies that were formed by multiple insulating layers stacked in a direction of stacking, in the length direction here.

(232) Firing the singulated chips turned the conductor pattern for coil conductors on and the conductor pattern for via conductors in the coil sheets into a coil conductor and a via conductor, respectively, thereby giving coils that were formed by multiple coil conductors stacked in the length direction and electrically coupled together by via conductors.

(233) In this way, bodies and coils disposed inside the bodies were produced. The direction of stacking of the insulating layers and the direction of the coil axis of the coil were parallel with the first primary surface, which was the mounting surface, of the body, parallel with the length direction here.

(234) Firing the singulated chips turned the conductor pattern for via conductors in the via sheets into via conductors, thereby producing first and second interlinking conductors that were formed by multiple via conductors stacked in the length direction and electrically coupled together. The first interlinking conductor became exposed on the first end surface of the body. The second interlinking conductor became exposed on the second end surface of the body.

(235) Then the corners and edges of the bodies were rounded by barrel-polishing with a medium in a rotary barrel.

(236) Outer Electrode Formation Step

(237) First, an electrically conductive paste containing Ag and glass frit was stretched to a predetermined thickness, and the bodies were dipped diagonally into the resulting layer. Then the resulting coating was baked at 810° C. to form an underlying electrode that extended from part of the first end surface of the body to part of each of the first primary surface, first lateral surface, and second lateral surface. Likewise, an underlying electrode was formed that extended from part of the second end surface of the body to part of each of the first primary surface, first lateral surface, and second lateral surface. The thickness of each underlying electrode was 5 μm .

(238) Then, by electrolytic plating, a Ni plating electrode and a Sn plating electrode were formed on each underlying electrode in sequence.

(239) In such a way, a first outer electrode, electrically coupled to the coil by the first interlinking

conductor, and a second outer electrode, electrically coupled to the coil by the second interlinking conductor, were formed.

(240) Through this, multilayer coil component samples 1 to 5 were produced. Multilayer coil component samples 1 to 5 each had a dimension in the length direction of 0.6 mm, a dimension in the height direction of 0.3 mm, and a dimension in the width direction of 0.3 mm.

(241) For multilayer coil component samples 1 to 5, when a first reference position is defined that coincides in the height direction with the end in the length direction of the portion of the underlying electrode in the first outer electrode lying on the first primary surface of the body, the first region in the first body section of the body was a combined range consisting of the range having a dimension in the length direction of 20 μm from the first reference position toward the second end surface (dimension P.sub.1 in FIG. 11) and the range having a dimension in the length direction of 20 μm from the first reference position toward the first end surface (dimension P.sub.2 in FIG. 11). For multilayer coil component samples 1 to 5, furthermore, the second region in the first body section was all but the first region.

(242) For multilayer coil component samples 1 to 5, when a second reference position is defined that coincides in the height direction with the end in the length direction of the portion of the underlying electrode in the second outer electrode lying on the first primary surface of the body, the first region in the second body section of the body was a combined range consisting of the range having a dimension in the length direction of 20 μm from the second reference position toward the first end surface (dimension Q.sub.1 in FIG. 12) and the range having a dimension in the length direction of 20 μm from the second reference position toward the second end surface (dimension Q.sub.2 in FIG. 12). For multilayer coil component samples 1 to 5, furthermore, the second region in the second body section was all but the first region.

(243) For multilayer coil component samples 1 to 5, the green sheets forming the first and second regions in the first and second body sections were combined as specified in Table 2. As a result, for multilayer coil component samples 1 to 5, the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases in the first region and the percentage by volume of the nonmagnetic phase to the total volume of the magnetic and nonmagnetic phases in the second region in the first and second body sections were as presented in Table 2. For multilayer coil component samples 1 to 5, furthermore, the Si content in terms of a SiO.sub.2 basis of the first region and the Si content in terms of a SiO.sub.2 basis of the second region in the first and second body sections were as presented in Table 2 (labeled as “SiO.sub.2 content” in Table 2).

(244) Testing

(245) Multilayer coil component samples 1 to 5 were tested as follows.

(246) Position of the Underlying and Plating Electrodes

(247) First, the multilayer coil component sample was sealed by filling its surroundings with resin, upright in such a manner that the first lateral surface was exposed upside. Then the multilayer coil component sample was polished using a grinder to substantially the middle in the width direction to expose a cross-section parallel with the length and height directions. After that, the exposed cross-section of the multilayer coil component sample underwent focused ion beam (FIB) milling using SII NanoTechnology's “SMI3050R” focused ion beam mill, giving a cross-section for observation.

(248) Then on the cross-section for observation of the multilayer coil component sample, the portion of the underlying and plating electrodes in the first outer electrode lying on the first primary surface of the body was imaged using a scanning electron microscope (SEM). Likewise, the portion of the underlying and plating electrodes in the second outer electrode lying on the first primary surface of the body was imaged. After that, the position of the underlying and plating electrodes was checked on the resulting images, with the result that in the length direction, the end of the plating electrode was closer than the end of the underlying electrode to the midpoint of the body. The distance in the length direction between the end of the plating electrode and the end of the underlying electrode, furthermore, was as presented in Table 2.

(249) TABLE-US-00002 TABLE 2 *Sample *Sample Sample Sample Sample 1 2 3 4 5 First Green sheets A B C C D region Nonmagnetic 0 50 60 60 100 phase (% by volume) SiO.sub.2 content — 23.9 31.4 31.4 82.3 (% by weight) Second Green sheets A A A B B region Nonmagnetic 0 0 0 50 50 phase (% by volume) SiO.sub.2 content — — — 23.9 23.9 (% by weight) Difference in SiO.sub.2 — 23.9 31.4 7.5 58.4 content between the first and second regions (% by weight) Distance in the length 40 50 22 20 15 direction between the end of the plating electrode and the end of the underlying electrode (μm)

(250) In Table 2, the samples having a name with an * are comparative examples, examples falling out of the scope of the present disclosure.

(251) As shown in Table 2, for multilayer coil component samples 3, 4, and 5, the distance in the length direction between the end of the plating electrode and the end of the underlying electrode was not longer than 30 μm.

(252) For multilayer coil component samples 1 and 2, on the other hand, the distance in the length direction between the end of the plating electrode and the end of the underlying electrode was longer than 30 μm.

(253) That is, for multilayer coil component samples 3, 4, and 5, compared with multilayer coil component samples 1 and 2, the plating electrode was prevented from outgrowing the underlying electrode along the first primary surface of the body.

(254) Transmission Coefficient S21

(255) With multilayer coil component samples 1, 3, and 4, the transmission coefficient S21, determined from the ratio of the power of the transmitted signal to that of the incident signal, was measured at varying frequencies from 30 GHz to 70 GHz using a network analyzer.

(256) FIG. 13 is a graphical representation of measured transmission coefficients S21 of multilayer coil component samples 1, 3, and 4.

(257) As shown in FIG. 13, multilayer coil component samples 3 and 4, compared with multilayer coil component sample 1, had a high transmission coefficient S21 in the radiofrequency band, for example at 40 GHz and 50 GHz, and were consistent in the transmission coefficient S21 in the radiofrequency and lower-frequency bands. Although not shown, multilayer coil component sample 5 was also found to have a high and consistent transmission coefficient S21 in the radiofrequency band like multilayer coil component samples 3 and 4.

Claims

1. A multilayer coil component comprising: a body including a plurality of insulating layers stacked in a direction of stacking and having first and second end surfaces opposite each other in a length direction, first and second primary surfaces opposite each other in a height direction, perpendicular to the length direction, and first and second lateral surfaces opposite each other in a width direction, perpendicular to the length direction and to the height direction; a coil disposed inside the body and configured by a plurality of coil conductors electrically connected together; and a first outer electrode extending from at least a portion of the first end surface of the body to a portion of the first primary surface and electrically connected to the coil, wherein: the direction of stacking of the insulating layers and a direction of a coil axis of the coil are parallel with the first primary surface, which is a mounting surface, of the body; at least a portion of the body has a magnetic phase containing Fe, Ni, Zn, and Cu and a nonmagnetic phase containing Si; the first outer electrode has, in order from a body side, an underlying electrode and a plating electrode on the underlying electrode; an end of the plating electrode lying in the length direction of a portion of the plating electrode lying on the first primary surface of the body is closer to the second end surface of the body than an end of the underlying electrode in the length direction of a portion of the underlying electrode lying on the first primary surface of the body; and a distance in the length direction between the end of the plating electrode and the end of the underlying electrode is 30 μm or less,

wherein: when a boundary is defined that extends parallel with the height and width directions at a midpoint of the body in the length direction, the body has first and second body sections aligned in the length direction with the boundary therebetween, the first body section including the first end surface and the second body section including the second end surface; when a first reference position is defined that coincides in the height direction with the end of the underlying electrode, the first body section has a first region including at least a range having a dimension of 20 μm to 60 μm in the length direction from the first reference position toward the second end surface; and the first region is totally the nonmagnetic phase or has both the nonmagnetic and magnetic phases.

2. The multilayer coil component according to claim 1, wherein the first region contains, in a total of 100% by weight, 30.0% by weight or more Si in terms of a SiO_2 basis.

3. The multilayer coil component according to claim 2, wherein the first region contains, in a total of 100% by weight, Si being from 30.0% by weight to 85.0% by weight in terms of a SiO_2 basis, B being from 4.0% by weight to 15.0% by weight in terms of a B_2O_3 basis, Fe being from 0% by weight to 45.0% by weight in terms of an Fe_2O_3 basis, Ni being from 0% by weight to 15.0% by weight in terms of a NiO basis, Zn being from 0% by weight to 8.0% by weight in terms of a ZnO basis, and Cu being from 0% by weight to 5.0% by weight in terms of a CuO basis.

4. The multilayer coil component according to claim 3, wherein the first region contains, in a total of 100% by weight, K being from 0.3% by weight to 1.5% by weight in terms of a K_2O basis and Mg being from 0.9% by weight to 3.5% by weight in terms of a MgO basis.

5. The multilayer coil component according to claim 4, wherein: the first body section further has a second region other than the first region; and the second region is totally the magnetic phase or has both the magnetic and nonmagnetic phases.

6. The multilayer coil component according to claim 5, wherein the second region contains, in a total of 100% by weight, 25.0% by weight or less Si in terms of a SiO_2 basis.

7. The multilayer coil component according to claim 5, wherein a Si content of the first region in a total of 100% by weight is larger than a Si content of the second region in a total of 100% by weight by 7.0% by weight or more in terms of a SiO_2 basis.

8. The multilayer coil component according to claim 3, wherein: the first body section further has a second region other than the first region; and the second region is totally the magnetic phase or has both the magnetic and nonmagnetic phases.

9. The multilayer coil component according to claim 8, wherein the second region contains, in a total of 100% by weight, 25.0% by weight or less Si in terms of a SiO_2 basis.

10. The multilayer coil component according to claim 8, wherein a Si content of the first region in a total of 100% by weight is larger than a Si content of the second region in a total of 100% by weight by 7.0% by weight or more in terms of a SiO_2 basis.

11. The multilayer coil component according to claim 2, wherein: the first body section further has a second region other than the first region; and the second region is totally the magnetic phase or has both the magnetic and nonmagnetic phases.

12. The multilayer coil component according to claim 11, wherein the second region contains, in a total of 100% by weight, 25.0% by weight or less Si in terms of a SiO_2 basis.

13. The multilayer coil component according to claim 11, wherein a Si content of the first region in a total of 100% by weight is larger than a Si content of the second region in a total of 100% by weight by 7.0% by weight or more in terms of a SiO_2 basis.

14. The multilayer coil component according to claim 1, wherein: the first body section further has a second region other than the first region; and the second region is totally the magnetic phase or has both the magnetic and nonmagnetic phases.

15. The multilayer coil component according to claim 14, wherein the second region contains, in a total of 100% by weight, 25.0% by weight or less Si in terms of a SiO_2 basis.

16. The multilayer coil component according to claim 15, wherein the second region contains, in a total of 100% by weight, Si being from 0% by weight to 25.0% by weight in terms of a SiO_2 basis,

B being from 0% by weight to 5.0% by weight in terms of a B₂O₃ basis, Fe being from 45.0% by weight to 70.0% by weight in terms of an Fe₂O₃ basis, Ni being from 10.0% by weight to 20.0% by weight in terms of a NiO basis, and Zn being from 5.0% by weight to 12.0% by weight in terms of a ZnO basis.

17. The multilayer coil component according to claim 16, wherein A Si content of the first region in a total of 100% by weight is larger than a Si content of the second region in a total of 100% by weight by 7.0% by weight or more in terms of a SiO₂ basis.

18. The multilayer coil component according to claim 15, wherein a Si content of the first region in a total of 100% by weight is larger than a Si content of the second region in a total of 100% by weight by 7.0% by weight or more in terms of a SiO₂ basis.

19. The multilayer coil component according to claim 14, wherein a Si content of the first region in a total of 100% by weight is larger than a Si content of the second region in a total of 100% by weight by 7.0% by weight or more in terms of a SiO₂ basis.
